

Chapter 36: Abstract Inner Product Spaces

In this section we use the properties of dot product on $\mathbb{R}^n$ for generalizing to the notion of the **inner product** to any real vector space.

Let's briefly review dot product first: For two vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ in

$\mathbb{R}^n$, the **dot product** is

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i$$

The **length**, or **magnitude**, or **norm** of the vector $\mathbf{x}$ is $\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$, and the distance between two vectors $\mathbf{x}$ and $\mathbf{y}$ is $\|\mathbf{x} - \mathbf{y}\|$.

Example: Find $\mathbf{x} \cdot \mathbf{y}$ for $\mathbf{x} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and find the norm of $\mathbf{x}$.

$$\mathbf{x} \cdot \mathbf{y} = 3 \times 2 + (-1) \times 1 = 5.$$

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{9 + 1} = \sqrt{10}.$$

Theorem: The following properties hold for dot product:

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
3. $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$ for k in $\mathbb{R}$.
4. $\mathbf{v} \cdot \mathbf{v} > 0$ for nonzero vector $\mathbf{v}$.

Example: Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$, and define

$$\mathbf{u} * \mathbf{v} = u_1 v_1 + 3u_2 v_2.$$

If we replace the dot product with this “new” product, do properties (1) through (4) hold?

(1): Does $\mathbf{u} * \mathbf{v} = \mathbf{v} * \mathbf{u}$?

$$\text{Yes.} \quad \mathbf{u} * \mathbf{v} = u_1 v_1 + 3u_2 v_2 = v_1 u_1 + 3v_2 u_2 = \mathbf{v} * \mathbf{u}$$

(2): Does $(\mathbf{u} + \mathbf{v}) * \mathbf{w} = \mathbf{u} * \mathbf{w} + \mathbf{v} * \mathbf{w}$?

$$\begin{aligned} \text{Yes.} \quad (\mathbf{u} + \mathbf{v}) * \mathbf{w} &= (u_1 + v_1)w_1 + 3(u_2 + v_2)w_2 \\ &= u_1 w_1 + 3u_2 w_2 + v_1 w_1 + 3v_2 w_2 \\ &= \mathbf{u} * \mathbf{w} + \mathbf{v} * \mathbf{w} \end{aligned}$$

(3): Does $(k\mathbf{u}) * \mathbf{v} = k(\mathbf{u} * \mathbf{v})$?

$$\begin{aligned} \text{Yes.} \quad (k\mathbf{u}) * \mathbf{v} &= (ku_1)v_1 + 3(ku_2)v_2 \\ &= k(u_1 v_1) + k \cdot 3(u_2 v_2) = k(\mathbf{u} * \mathbf{v}) \end{aligned}$$

(4): For nonzero $\mathbf{v}$, is $\mathbf{v} * \mathbf{v} > 0$?

$$\text{Yes.} \quad \mathbf{v} * \mathbf{v} = v_1^2 + 3v_2^2 > 0 \text{ for nonzero } \mathbf{v}.$$

What is the “norm” of $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$?

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} * \mathbf{u}} = \sqrt{9 + 3} = \sqrt{12}$$

Definition: Let V be a vector space. A (real) **inner product** on V is a function that assigns to each **ordered pair** of vectors $\mathbf{v}, \mathbf{w}$ in V a **real number** $\langle \mathbf{v}, \mathbf{w} \rangle$ satisfying the following properties:

- (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v}$ in V .
- (b) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
- (c) $\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{u}$ and $\mathbf{v}$ in V and k in $\mathbb{R}$.
- (d) $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in V .

V with $\langle \cdot, \cdot \rangle$ is called an **inner product space**.

Example: For $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$, the function

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 3u_2 v_2$$

is an inner product on $\mathbb{R}^2$.

Example: Let V be the vector space of all continuous real-valued functions on the interval $[0, 1]$ (If this bothers you, replace V with P = set of all polynomials).

Let

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx.$$

Is it an inner product on V ?

Yes. (a). $\langle f, g \rangle = \langle g, f \rangle$

(b). $\langle f+g, h \rangle = \int (f+g)h = \int fh + \int gh$
 $= \langle f, h \rangle + \langle g, h \rangle$

(c). $\langle kf, g \rangle = \int kfg = k \int fg = k \langle f, g \rangle$

(d) $f \neq 0, \langle f, f \rangle = \int f^2 > 0$.

Theorem: For an inner product $\langle \cdot, \cdot \rangle$ on a vector space V , the following additional properties automatically hold:

1. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$ for any $\mathbf{u}, \mathbf{v}$ in V .
2. $\langle \mathbf{u}, k\mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$ for any $\mathbf{u}, \mathbf{v}$ in V , and k in $\mathbb{R}$.
3. $\langle \mathbf{v}, \mathbf{0} \rangle = 0 = \langle \mathbf{0}, \mathbf{v} \rangle$
4. $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Example: The dot product on $\mathbb{R}^n$ is an inner product. We will often call the dot product the **standard inner product**.

Example: Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 3u_2 v_2$$

Show this satisfies property (d): $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ for $\mathbf{u} \neq \mathbf{0}$.

$$\langle \mathbf{u}, \mathbf{u} \rangle = u_1^2 - 2u_1 u_2 + 3u_2^2 = (u_1 - u_2)^2 + 2u_2^2 > 0$$

for $\mathbf{u} \neq \mathbf{0}$.

Theorem: Let $E = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$, and let $\langle \cdot, \cdot \rangle$ be an inner product on $\mathbb{R}^n$. Let $c_{ij} = \langle \mathbf{e}_i, \mathbf{e}_j \rangle$ and $C = [c_{ij}]$. Then

(a) C is a metric matrix.

(b) $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T C \mathbf{v}$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$.

Example: For the inner product on $\mathbb{R}^2$ defined by $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 3u_2 v_2$, find C such that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T C \mathbf{v}$.

$$C = \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$$

Question: Under what conditions on a $n \times n$ matrix A will $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T A \mathbf{v}$ form an inner product on $\mathbb{R}^n$?

A is a positive symmetric matrix.

Definition: A symmetric real $n \times n$ matrix A that satisfies $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ is called a **positive definite real matrix**.

Example: Consider M_{22} , and let A, B be matrices in M_{22} . Let

$$\langle A, B \rangle = \text{tr}(AB^T).$$

(note: your homework states $\text{tr}(AB^T)$ as $(AB^T)_{1,1} + (AB^T)_{2,2}$).
You can show this is an inner product.

Find $\langle A, B \rangle$ for $A = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}$.

$$\begin{aligned} \langle A, B \rangle &= \text{tr}(AB^T) = \text{tr}\left(\begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 \\ 2 & 1 \end{bmatrix}\right) \\ &= \text{tr}\left(\begin{bmatrix} 4 & 1 \\ -6 & -5 \end{bmatrix}\right) = -1. \end{aligned}$$

Norm and Distance

Definition: In an inner product space the **norm** of a vector $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Definition: The **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Example: Consider P_1 with the following inner product: $\langle p, q \rangle = \int_0^1 p(x)q(x)dx$.

What's the norm of $p(x) = x$?

$$\|p(x)\| = \sqrt{\langle x, x \rangle} = \sqrt{\int_0^1 x^2 dx} = \sqrt{\frac{1}{3}}$$

Find the distance between $q_1(x) = 2x + 1$ and $q_2(x) = 3x$.

$$\begin{aligned} \|q_1 - q_2\| &= \sqrt{\langle q_1 - q_2, q_1 - q_2 \rangle} \\ &= \int_0^1 (x-1)^2 dx = \left. \frac{(x-1)^3}{3} \right|_0^1 = \frac{1}{3}. \end{aligned}$$

Example: Consider M_{22} , with the inner product $\langle A, B \rangle = \text{tr}(AB^T)$. Find the distance between $A = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$.

$$\begin{aligned} \|A - B\| &= \sqrt{\langle A - B, A - B \rangle} = \sqrt{\text{tr}((A - B)(A - B)^T)} \\ &= \sqrt{\text{tr}\left(\begin{bmatrix} 4 & 1 \\ -1 & -3 \end{bmatrix} \cdot \begin{bmatrix} 4 & -1 \\ 1 & -3 \end{bmatrix}\right)} = \sqrt{17 + 10} \\ &= \sqrt{27}. \end{aligned}$$

Unit Vectors

Definition: A vector $\mathbf{v}$ in an inner product space V is called a **unit vector** if $\|\mathbf{v}\| = 1$.
(Note: This is equivalent to $\langle \mathbf{v}, \mathbf{v} \rangle = 1$).

Theorem: If $\mathbf{v}$ is a nonzero vector of an inner product space, then $\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$ is the unique unit vector that is a positive multiple of $\mathbf{v}$

Remark: We often say that $\mathbf{u}$ is “in the same direction as $\mathbf{v}$.”

Example: Consider P_1 with the following inner product: $\langle p, q \rangle = \int_0^1 p(x)q(x)dx$.

Find the unit vector in the same direction as $p(x) = x$.

$$\|p(x)\| = \sqrt{\langle x, x \rangle} = \sqrt{\frac{1}{3}}.$$

$$\frac{p}{\|p\|} = \sqrt{3}x.$$

Cauchy–Bunyakovsky–Schwarz (CBS) Inequality

If $\mathbf{u}, \mathbf{v}$ are vectors in an inner product space, then

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|.$$

Moreover, equality occurs if and only if one of $\mathbf{v}$ and $\mathbf{w}$ is a scalar multiple of the other.

Proof:

$$\text{Let } p(t) = \langle \mathbf{v}, t\mathbf{w} \rangle, \quad t \in \mathbb{R}.$$

$$\text{then } \langle \mathbf{v} + t\mathbf{w}, \mathbf{v} + t\mathbf{w} \rangle \geq 0 \text{ for } \forall t \in \mathbb{R}.$$

$$\Rightarrow \|\mathbf{w}\|^2 t^2 + 2\langle \mathbf{v}, \mathbf{w} \rangle t + \|\mathbf{v}\|^2 \geq 0, \text{ for } \forall t \in \mathbb{R}.$$

$$\Rightarrow \Delta = 4\langle \mathbf{v}, \mathbf{w} \rangle^2 - 4\|\mathbf{v}\|^2 \|\mathbf{w}\|^2 \leq 0$$

$$\Rightarrow |\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|.$$

Example: If $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are real numbers, then

$$\left| \sum_{i=1}^n a_i b_i \right| \leq \left(\sqrt{\sum_{i=1}^n a_i^2} \right) \left(\sqrt{\sum_{i=1}^n b_i^2} \right).$$

Triangle Inequality

Theorem: If $\mathbf{v}$ and $\mathbf{w}$ are vectors in an inner product space, then

$$\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|.$$

Proof: (if time):

$$\begin{aligned} \|\mathbf{v} + \mathbf{w}\|^2 &= \langle \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} \rangle \\ &= \|\mathbf{v}\|^2 + 2\langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{w}\|^2 \\ &\leq \|\mathbf{v}\|^2 + 2\|\mathbf{v}\| \cdot \|\mathbf{w}\| + \|\mathbf{w}\|^2 \\ &= (\|\mathbf{v}\| + \|\mathbf{w}\|)^2 \\ \Rightarrow \|\mathbf{v} + \mathbf{w}\| &\leq \|\mathbf{v}\| + \|\mathbf{w}\|. \end{aligned}$$

Theorem: If V is an inner product space, the norm $\|\cdot\|$ has the following properties:

1. $\|\mathbf{v}\| \geq 0$ for every vector $\mathbf{v}$ in V .
2. $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
3. $\|k\mathbf{v}\| = |k|\|\mathbf{v}\|$ for every $\mathbf{v}$ in V and every k in $\mathbb{R}$.
4. $\|\mathbf{v} - \mathbf{w}\| = \|\mathbf{w} - \mathbf{v}\|$ for all $\mathbf{v}, \mathbf{w}$ in V .

$$1. \text{ Since } \|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} \geq 0, \forall \mathbf{v} \in V.$$

$$2. \|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = 0 \text{ iff } \mathbf{v} = \mathbf{0}.$$

$$3. \|k\mathbf{v}\| = \sqrt{\langle k\mathbf{v}, k\mathbf{v} \rangle} = \sqrt{k^2 \langle \mathbf{v}, \mathbf{v} \rangle} = |k| \cdot \|\mathbf{v}\|$$

$$4. \|\mathbf{v} - \mathbf{w}\| = \sqrt{\langle \mathbf{v} - \mathbf{w}, \mathbf{v} - \mathbf{w} \rangle} \stackrel{37}{=} \sqrt{\langle \mathbf{w} - \mathbf{v}, \mathbf{w} - \mathbf{v} \rangle} = \|\mathbf{w} - \mathbf{v}\|.$$